Narnarayan Shastri Institute of Technology
&
ECLearnix EdTech Pvt. Ltd. Organizes
International Level Hackathon 360° - 4.O
Round 1

Chosen Hackathon Domain :

- Domain 1 - Open challenge Solutions (Project,case study etc)
- Domain 2 - Business Solutions
- Domain 3 - EdTech Solutions
- Domain 4 - IT Solutions
- Domain 5 - Cyber Security & Forensic Science
- Domain 6 - Healthcare & Allied

Team Details :

- Team Name : SyntaxError
- Domain Name : Open challenge Solutions
- Single Point of Contact (SPOC) or Team lead Name : Anit Sarkar
- SPOC Email ID : anitsarkar04@gmail.com
- SPOC Mobile Number :8777547554

TEAM MEMBER'S NAMES & CONTACT DETAILS

S.No	Team Member Name (*)	Hackathon ID (Visit Hackathon Portal in our website)	WhatsApp Contact Number	Role in the Hackathon (#)	Designation(Employee/ UG/ PG Student / etc)	College / Organization Name and Department State & Country
1.	ANIT SARKAR	ECLIH40458	8777547554	**SPOC and Developer**	UG Student	Academy Of Technology
2.	ANINDITA MAZUMDAR	ECLIH40468	9083867886	Documentation	UG Student	Academy Of Technology
3.	ARPAN BHATTACHARYA	ECLIH40481	7478338242	Tester	UG Student	Academy Of Technology
4.	DIPMALLYA GOSWAMI	ECLIH40466	9330068254	Presentation	UG Student	Academy Of Technology
5.						
6.						

(#) - Project Leader, Research Analyst, Developer, Tester / Debugger, Documentation Lead , Presentation Lead, etc..

Problem Statements

Domain 1 – Open Challenges - Identify and solve a real-world problem with an innovative, practical, and scalable solution aligned to any domain.

Domain 2 – Business & EdTech Solutions: Improve user engagement, retention, and business value of an existing learning app through small, high-impact enhancements without redesigning it.

Domain 3 – IT Solutions: Optimize pre-launch user experience and engagement of an educational events platform to ensure strong adoption and retention..

Domain 4 – IT Solutions: Automate the conversion of raw DOCX manuscripts into publication-ready formats using AI with minimal manual effort.

Domain 5 – Cyber Security & Forensic Science: Build an AI-powered system to detect and prevent cyber threats like fake registrations, unauthorized access, and data breaches.

Domain 6 - Healthcare & Allied: Identify a real-life healthcare problem and develop a simple, innovative solution to improve accessibility, awareness, or management.

Abstract & Domain Name

- This project presents an AI-based music generation system designed to automate the creation of songs from simple user prompts. Traditional music production requires significant time, expertise, and resources, which limits accessibility for content creators and developers. The proposed system addresses this problem by leveraging artificial intelligence models to generate lyrics, audio, and visual assets automatically.
- The system follows a scalable pipeline architecture where user input is processed and added to a task queue. The backend, deployed on serverless GPU infrastructure, handles music generation using pre-trained AI models. The system integrates multiple components including lyric generation using language models, music generation using diffusion-based models, and thumbnail generation for visual representation. Generated outputs are stored and made accessible to users through a web interface.
- The objective of this project is to provide a fast, scalable, and user-friendly solution for automated music creation. The system supports multiple modes such as automatic generation, manual input control, and instrumental generation, allowing flexibility for different use cases.
- The expected outcome is a reliable AI-powered platform that reduces the time and effort required for music production, enabling creators to generate customized music efficiently. This solution contributes to innovation in digital content creation and demonstrates the practical application of AI in creative industries.So this Project Clearly fits on Industary,Innovation and Infrastructure

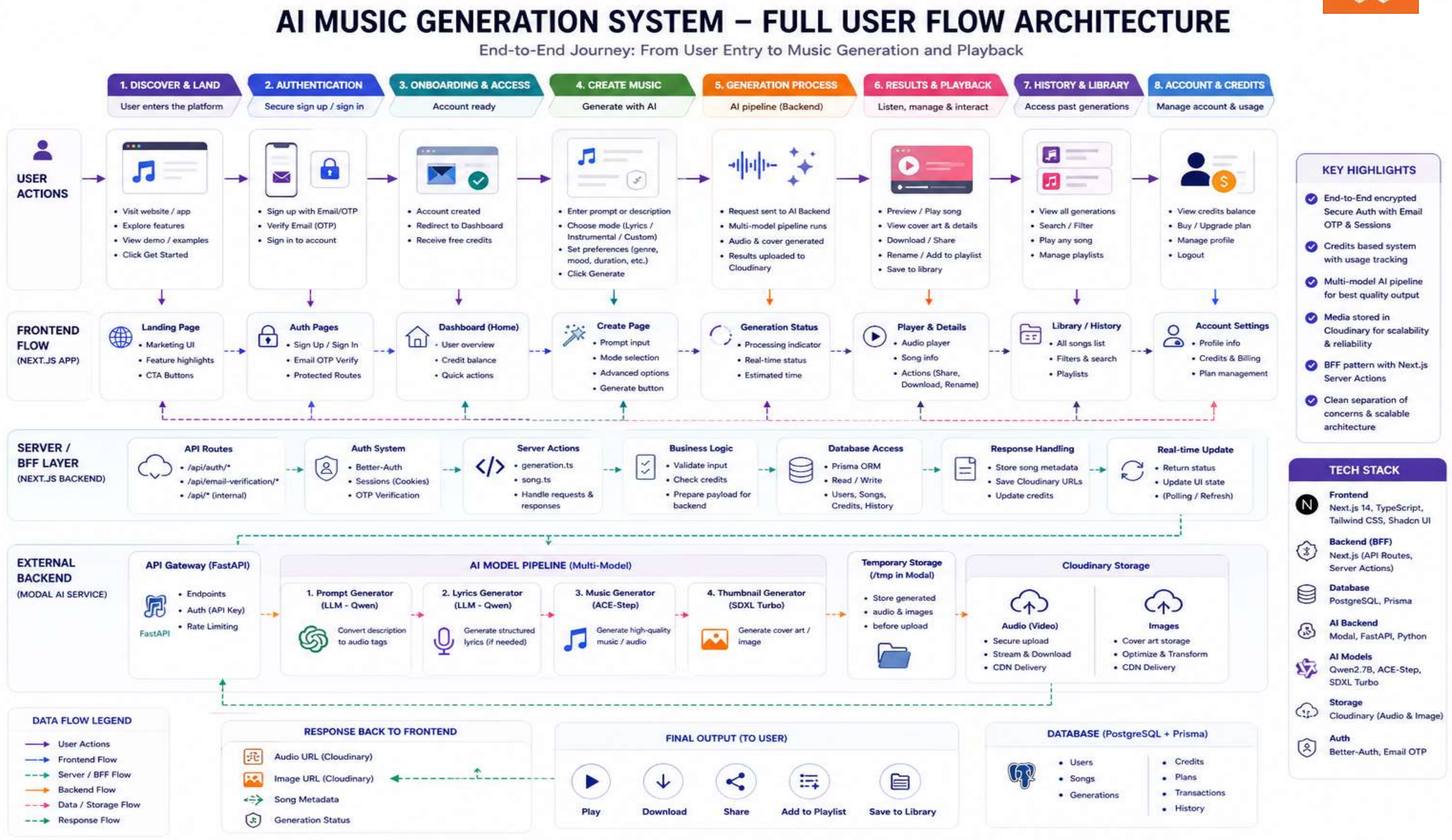
Objective of the Project

- Develop an end-to-end AI-powered system capable of automatically generating complete music tracks—including lyrics, audio composition, and visual assets—from simple user inputs, eliminating the need for manual music production workflows.
- Minimize the time required for music creation by replacing traditional multi-step production processes with a fully automated pipeline, enabling users to generate high-quality music within minutes instead of hours or days.Provide scalable backend system for generation
- Design and implement a scalable backend architecture using serverless GPU infrastructure to efficiently handle multiple concurrent generation requests, ensuring reliable performance and optimized resource utilization.
- Build an intuitive and accessible web-based interface that allows users with no prior musical or technical expertise to generate, manage, and interact with AI-generated music seamlessly.
- Provide flexible music generation options, including automatic generation, custom lyrics input, and instrumental mode, allowing users to control the creative process based on their specific requirements.

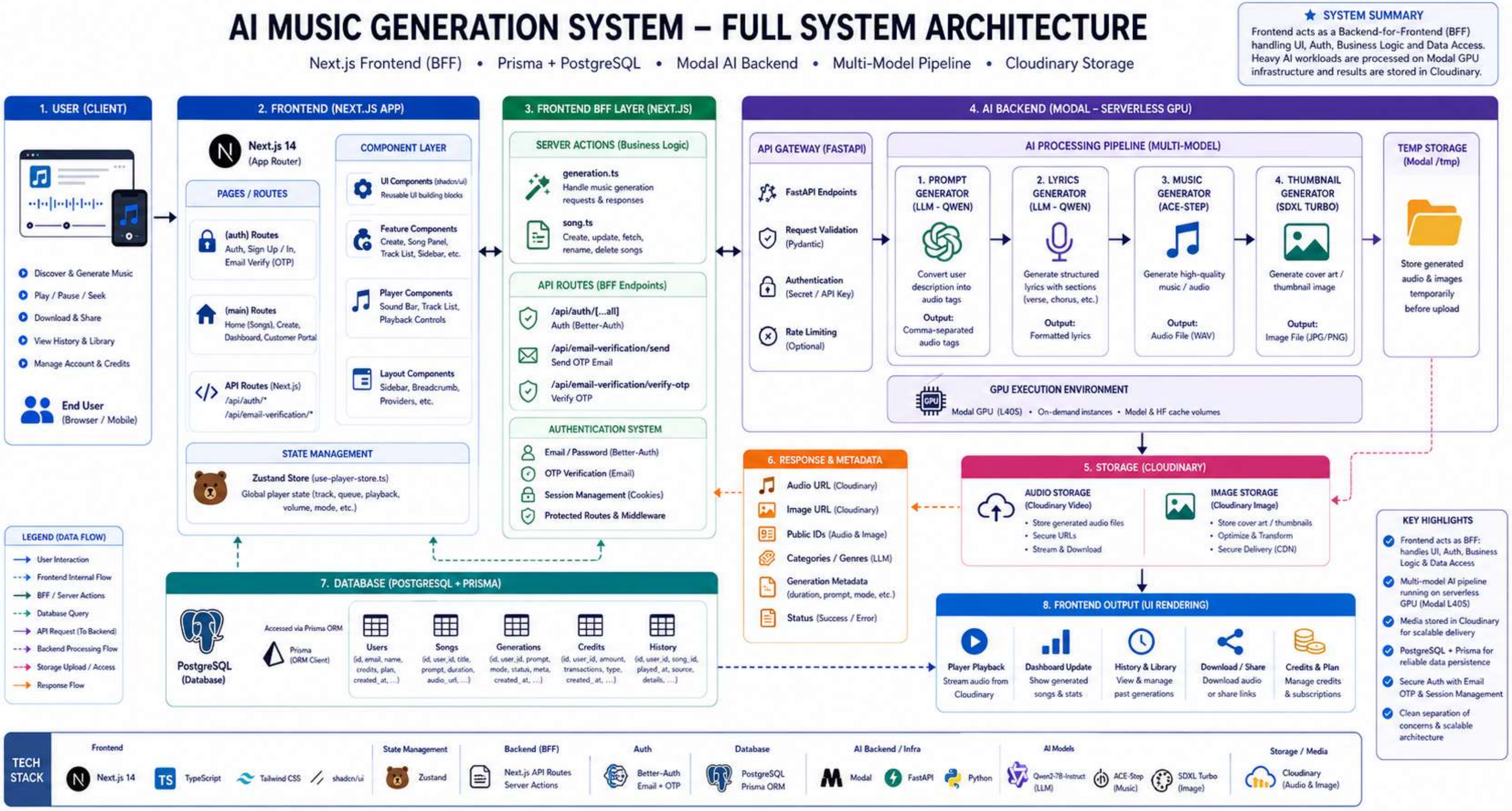
Proposed System

- The proposed system is a full-stack AI-powered music generation platform that enables users to create complete songs from simple textual prompts. The system follows a modular and scalable architecture combining frontend interaction, backend orchestration, and AI-driven content generation.
- The frontend is built using Next.js with an App Router architecture, providing a responsive and component-based user interface. It includes features such as music creation forms, a global audio player, dashboard views, and history management. A state management system ensures seamless playback and user interaction.
- The system uses a Backend-for-Frontend (BFF) layer implemented via Next.js server actions and API routes, which handles authentication, business logic, and communication with external services. A secure authentication mechanism using email and OTP ensures user-specific access and session management.
- The backend processing is handled by a serverless GPU-based system deployed on Modal. This backend executes a multi-stage AI pipeline that includes prompt generation, lyric generation, music composition, and thumbnail generation using multiple pre-trained models.
- Generated media files are temporarily processed and then stored using a cloud-based media storage system, ensuring efficient delivery and scalability. Metadata such as user data, generated songs, and usage credits are managed using a database system.
- Overall, the system provides a complete end-to-end pipeline from user input to music generation, storage, and playback, ensuring scalability, efficiency, and usability.

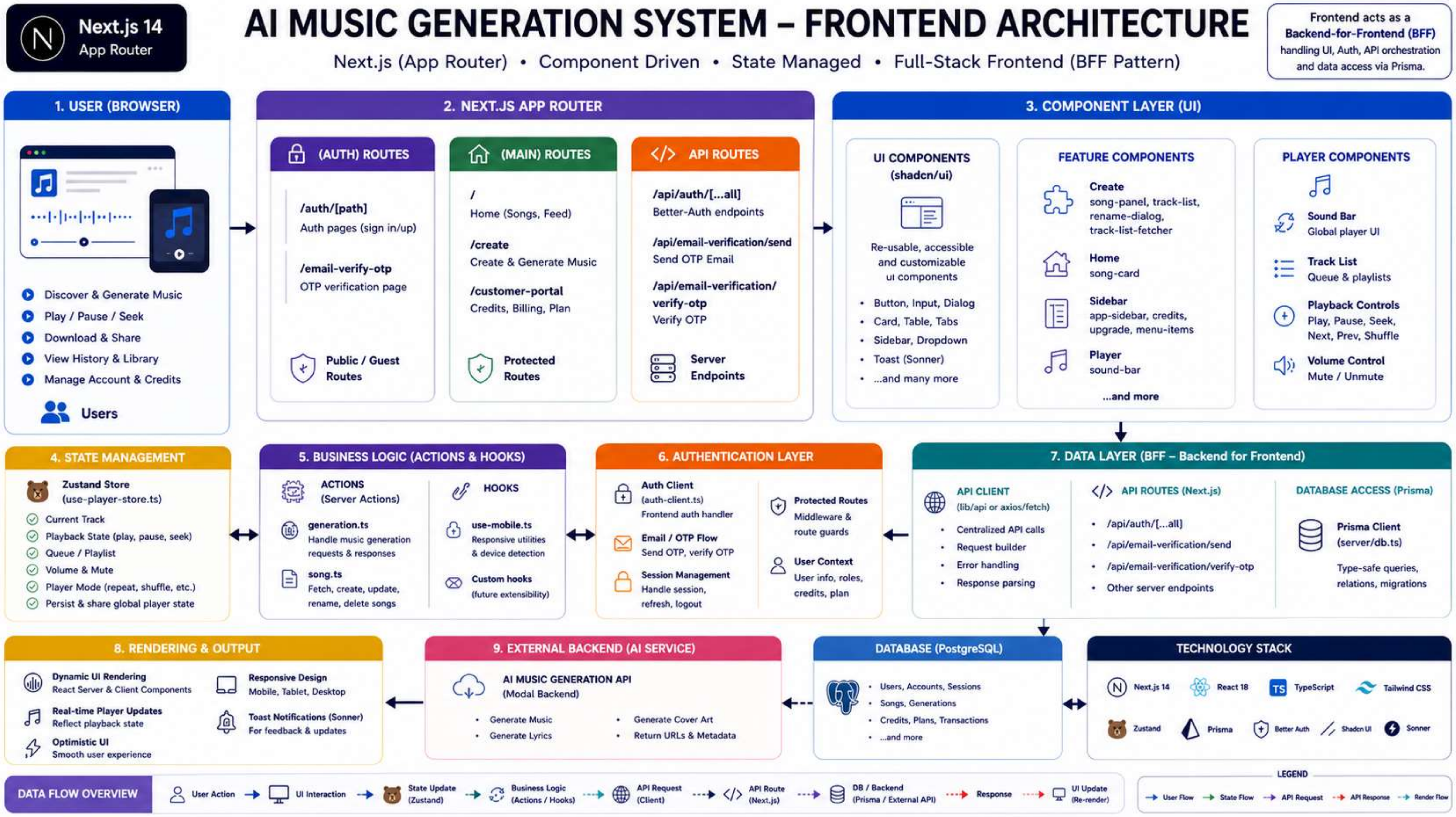
Methodology / Experimental Flow



Methodology / Experimental Flow



Methodology / Experimental Flow



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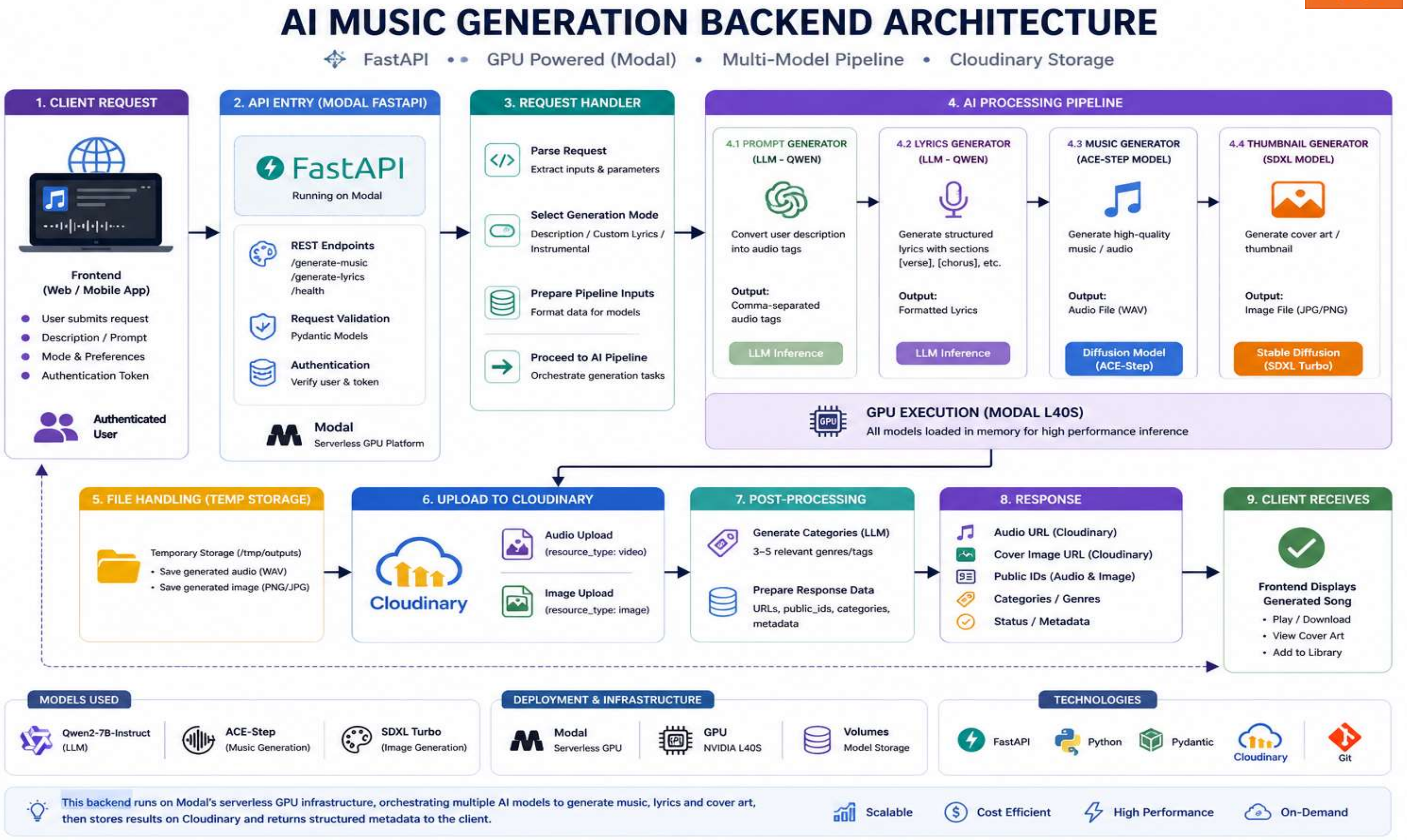

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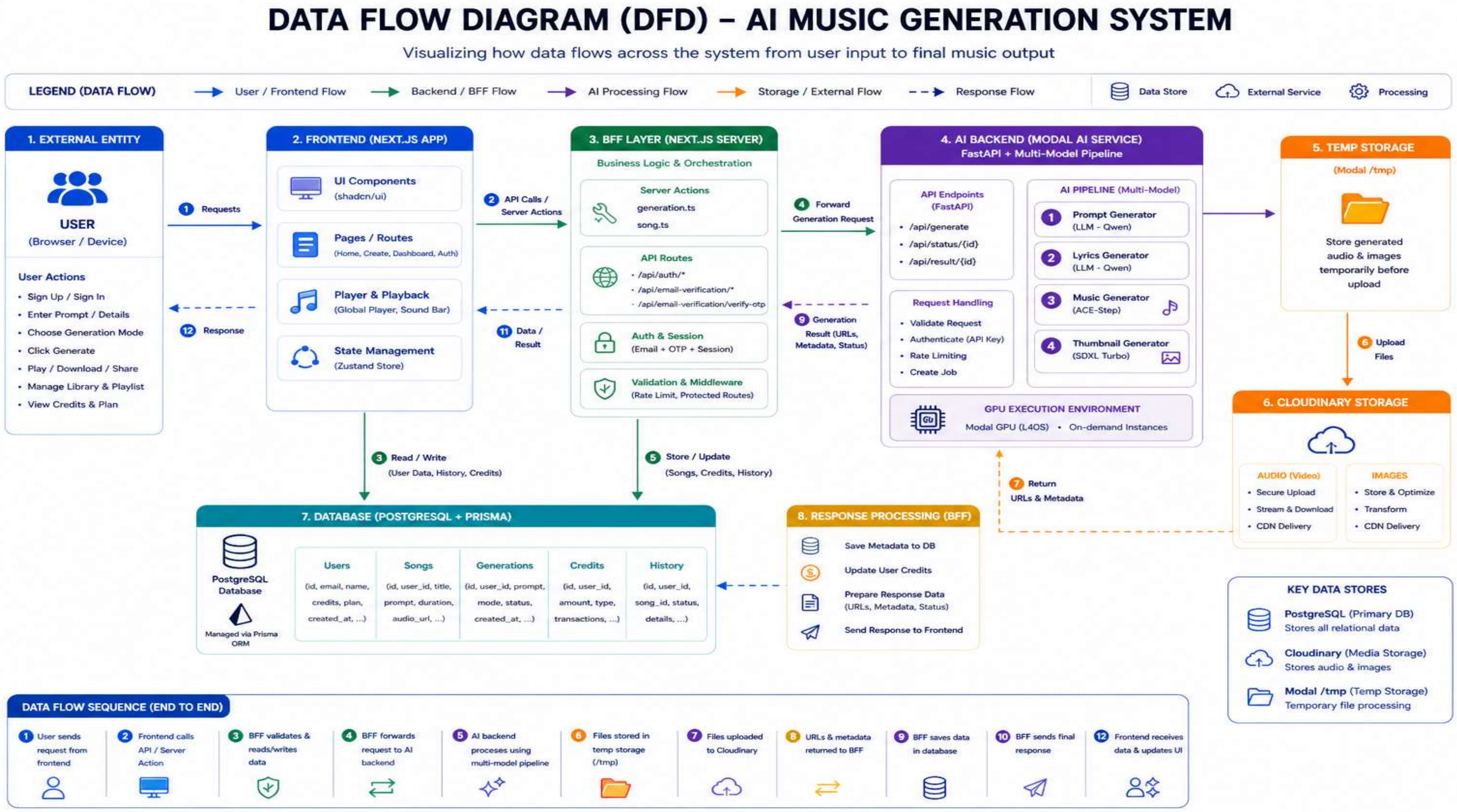

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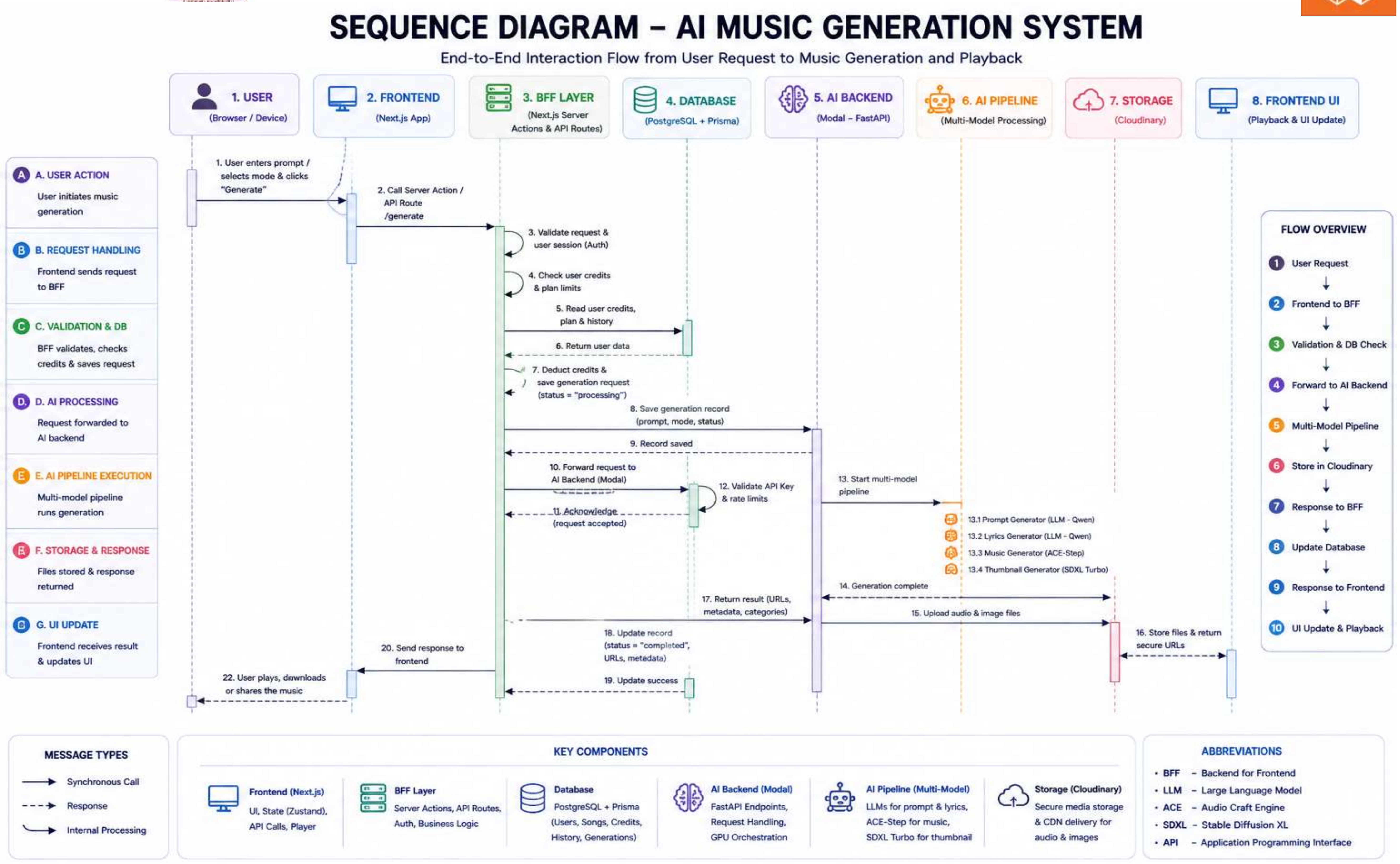

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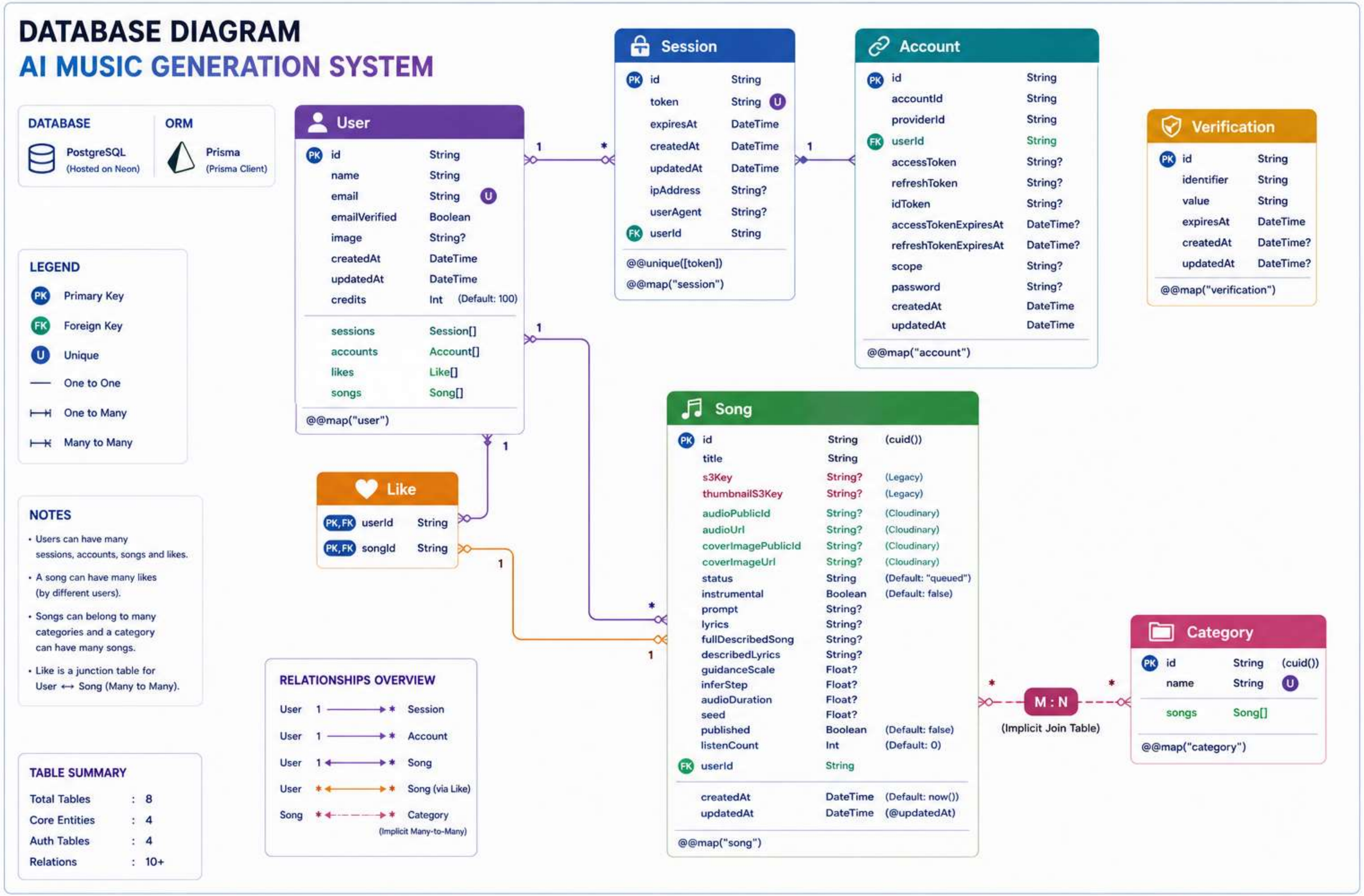

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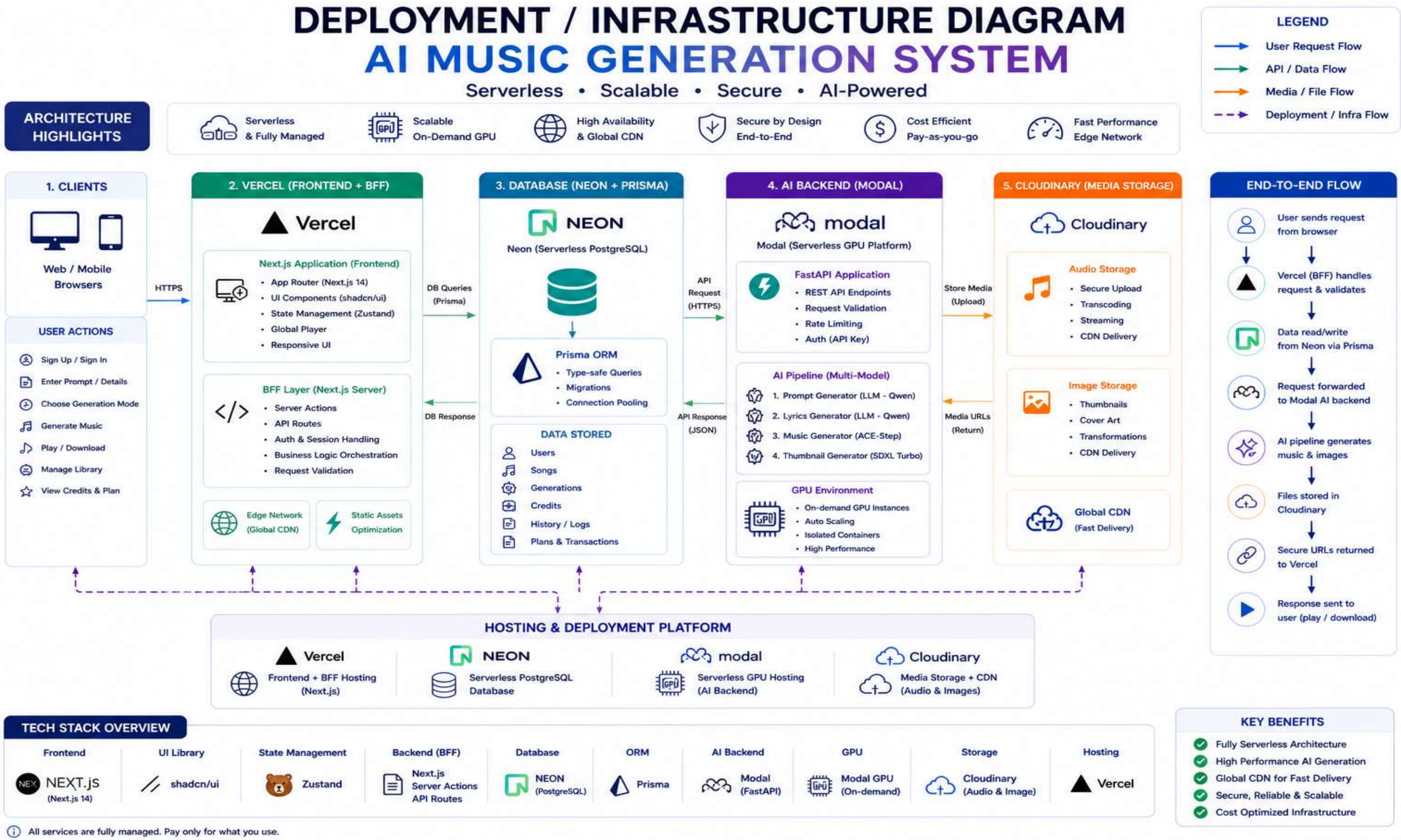


Results & Observations

- The system was tested with multiple user prompts and generation scenarios, demonstrating successful end-to-end functionality. The AI pipeline was able to generate lyrics, music, and thumbnails consistently based on user input.
- The integration of multiple AI models worked effectively, producing coherent outputs across different components of the system. The backend architecture handled requests efficiently, and the serverless GPU execution ensured that resource-intensive tasks were processed without blocking the user interface.
- The frontend system successfully rendered generated content in real time, allowing users to play, download, and manage their music seamlessly. The global player state ensured smooth playback across different pages.
- The authentication system functioned reliably, enabling secure user access and personalized content management. The database integration allowed proper storage and retrieval of user-generated content and metadata.

Overall, the system achieved stable performance, demonstrating the feasibility of integrating AI-driven content generation into a scalable web platform.

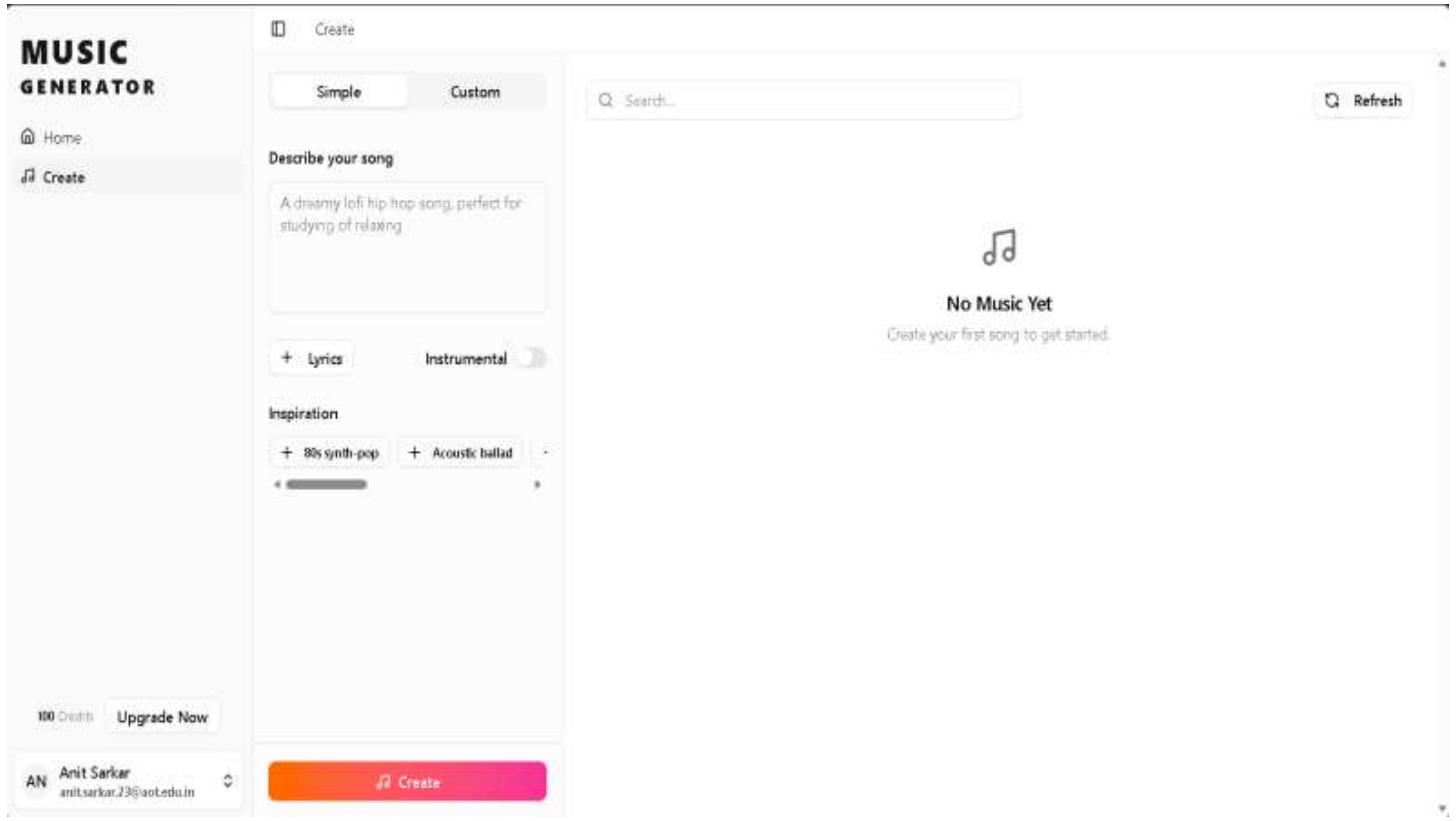
Expected Impact / Benefits

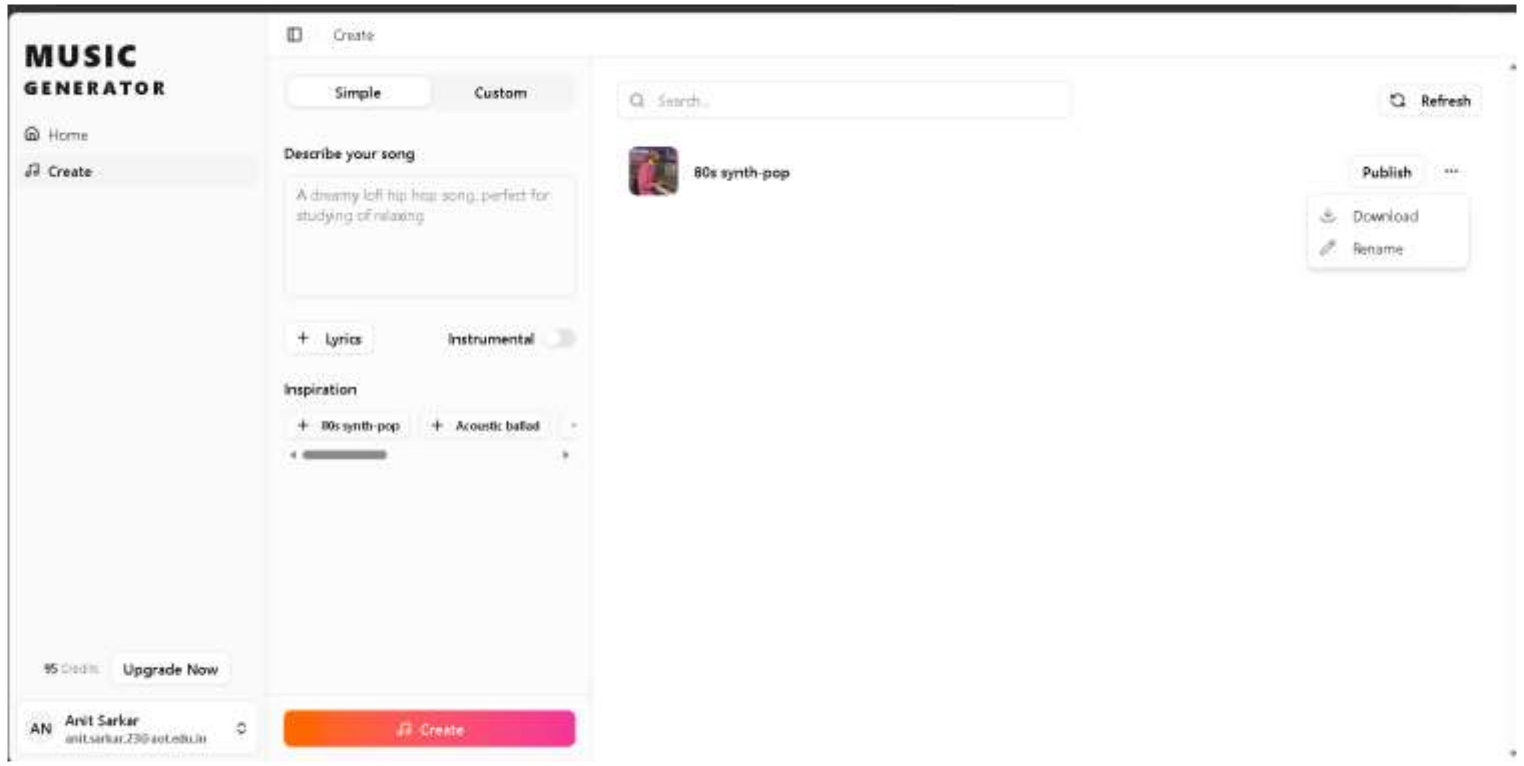
- The proposed system has significant potential to transform the way music is created and consumed by lowering the barrier to entry for music production. It enables individuals without technical or musical expertise to generate high-quality music using simple inputs.
- This solution can benefit content creators, developers, and digital media platforms by providing a fast and cost-effective way to generate custom music for various use cases such as videos, games, and online content.
- The system also demonstrates the practical application of artificial intelligence in creative industries, highlighting how multiple AI models can be integrated into a unified pipeline for real-world use.
- By reducing time, cost, and complexity, the platform contributes to increased productivity and innovation in digital content creation.
- Additionally, the scalable architecture ensures that the system can be extended and deployed for larger audiences, making it suitable for commercial and industrial applications.

Future Enhancements & Challenges Faced

- **Challenges:**
- High GPU computation cost
- Model latency
- Quality consistency
- Data dependency
- **Future:**
- Improve generation quality
- Reduce processing time
- Add more customization
- Enhance scalability
- Give an API to Intergrade with another apps

Images


MUSIC GENERATOR
Home
Create
Simple
Custom
Describe your song
A dreamy lofi hip hop song, perfect for studying of relaxing
Lyrics
Instrumental
Inspiration
80s synth-pop
Acoustic ballad
Search...
Refresh
No Music Yet
Create your first song to get started
100 Credits
Upgrade Now
AN
Anit Sarkar
anit.sarkar.23@aot.edu.in
Create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Publish
Download
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95 Credits
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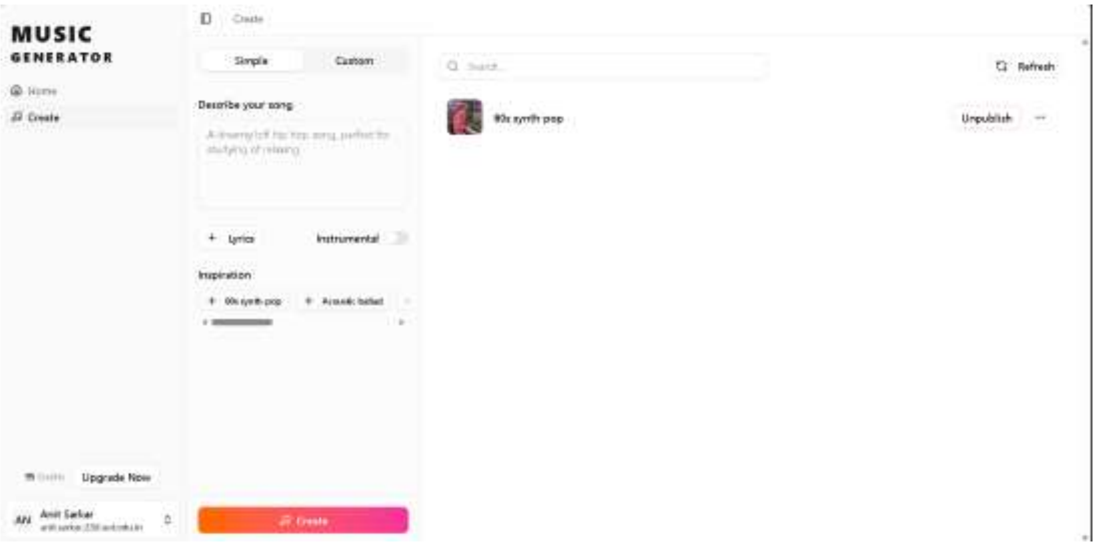
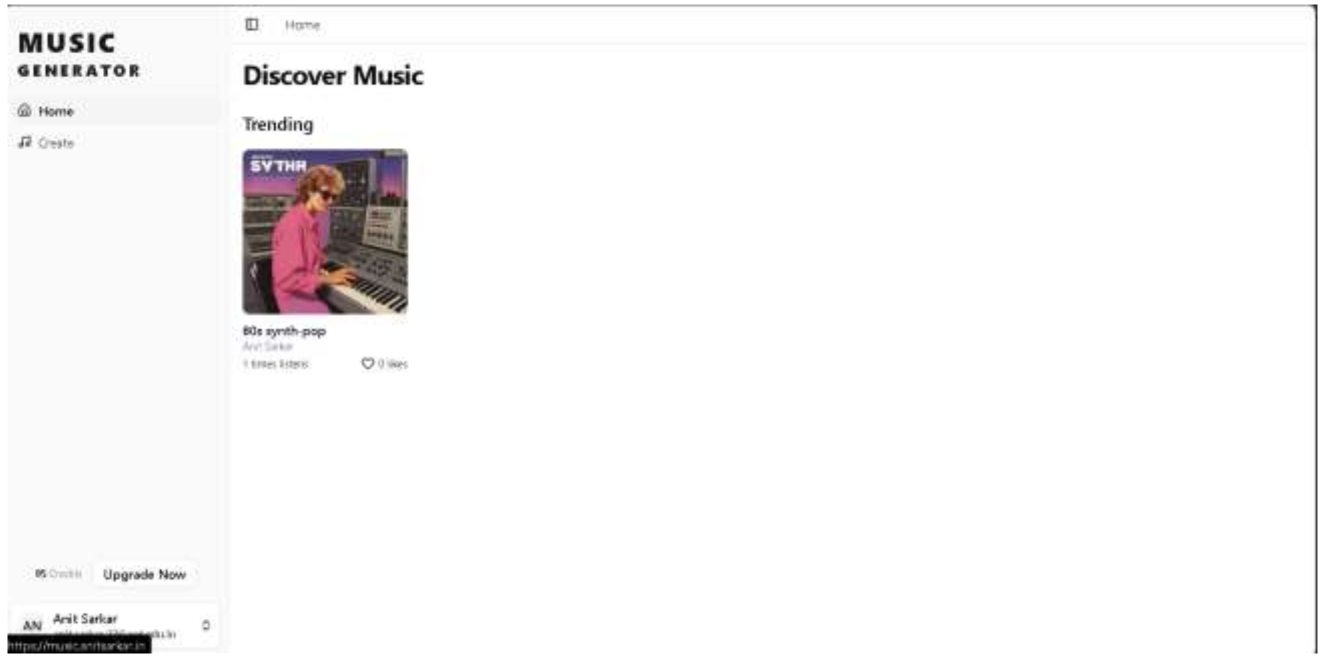
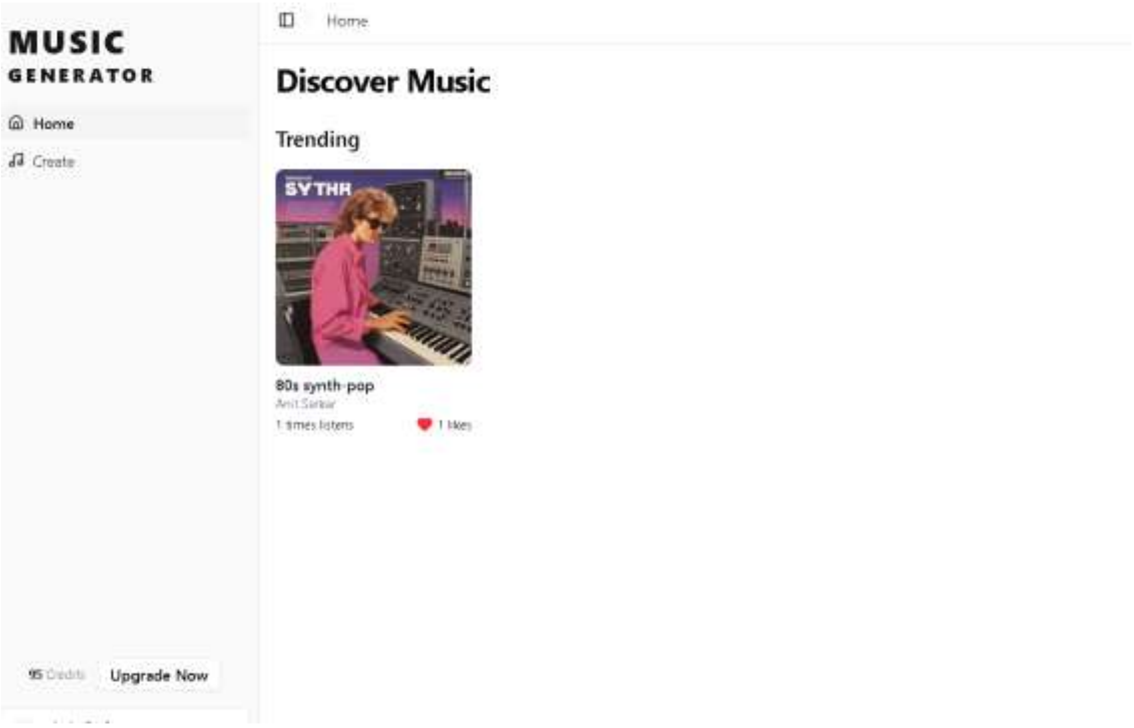
AN
Anit Sarkar
Create


MUSIC GENERATOR
Simple
Custom
Describe your song
Instrumental
Inspiration
Refresh
Unpublish
Upgrade Now
Create


MUSIC GENERATOR
Home
Create
Discover Music
Trending
80s synth-pop
Upgrade Now
Anit Sarkar


MUSIC GENERATOR
Home
Create
Discover Music
Trending
80s synth-pop
Anit Sarkar
1 times listens
1 likes
95 Credits
Upgrade Now

Conclusion & Thank You

- In conclusion, this project presents a complete AI-based music generation system that integrates frontend interaction, backend processing, and multi-model AI pipelines into a unified platform.
- The system successfully automates the process of music creation, reducing manual effort and production time while maintaining usability and scalability. The use of a Backend-for-Frontend architecture, serverless GPU computing, and cloud-based storage ensures efficient performance and modular design.
- The project demonstrates how AI can be effectively applied in creative domains to build practical and scalable solutions. With further improvements in model quality and system optimization, this platform has the potential to evolve into a fully production-ready system for real-world applications.


Thank
you